ADAPTIVE FILTERS USING IIR AND LAGUERRE STRUCTURE FOR SYSTEM IDENTIFICATION

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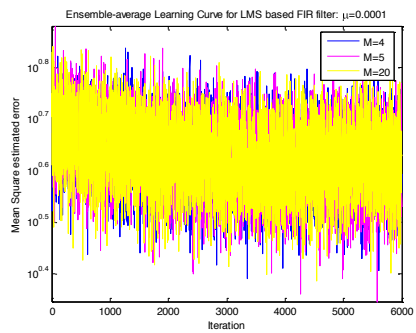
Abstract: The concept of IIR adaptive filter is reviewed in this paper. Both the unstrained IIR and Laguerre lattice structure are detailed. The numerical analysis based on output error method, equation error method and the gradient-adaptive laguerre algorithm are performed for the direct IIR and the Lattice form laguerre filter. Simulation results show the advantage of IIR filter over FIR in some aspects, on the other hand they also indicate that more advanced techniques need to be developed to overcome the inherent drawbacks of the IIR filters such as instability, local optimum, low convergence rate, sensitive to initial condition and coefficient etc.

1. INTRODUCTION

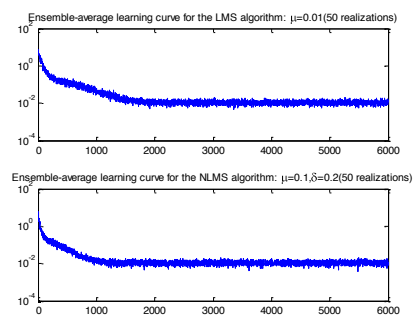
IIR filters are generally applied in the cases where very long impulse responses are required to achieve a prescribed performance. Examples include echo canceling, voice coding, adaptive equalization, active noise control and signal enhancing, adaptive notch filtering. In such practical applications the IIR filters are more realizable due to the fact that they have less degree of orders and higher accuracy over the FIR filters. It is an interesting and challenging topic to develop effective algorithms for IIR filtering. Active research on this subject, however, results on adaptive IIR are not available for all classical applications. Different to FIR, IIR filters use specific modeling aspects of the application. There are no general purpose optimal algorithm exists. The primary objective of this paper is to present the characteristics of some classical algorithms for IIR adaptive filtering while applied to system identification applications.

The performance differences between the FIR and the IIR filter can be seen by a system identification example. The unknown system identified is a second order ARMA process. The performance of the FIR with order 4, 5, 20 are presented in Fig.1 by means of the ensemble average curves. Fig.2 shows the IIR filter results. The mean values and the standard deviations (Std) are also given. As we can see, the estimation errors of FIR filters are quite larger compare to that of the IIR filters. It indicates that we can not increase arbitrarily the filter length to improve modeling capabilities (due to misadjustment and complexity). For instance, in this example the FIR filter with order 20 still can not have apparent improvement in its performance. The comparison of the FIR and IIR results suggests the accuracy of the FIR filter may not meet the prescribed requirement and we may seek for solutions by using IIR filters for this particular application.



Filter order	Mean	Std
4	4.323	0.623
5	4.325	0.638
20	4.322	0.653

Fig.1 FIR performance for an ARMA identification



Algorithm	Mean	Std
LMS	0.074	0.355
NLMS	0.038	0.188

Fig.2 IIR performance for an ARMA identification

The paper is organized as follows. Section 2 discusses the structure of the IIR filters. In Section 3, several common algorithms including the output error (OE), equation error (EE) and the gradient-adaptive Laguerre Lattice (GALL) are detailed. The advantage/disadvantages of each method are also emphasized. In Section 4, the simulation analysis is presented followed by conclusions in Section 5.

2. STRUCTURE OF IIR FILTERS

Numerically an IIR filter consists of an impulse transfer function $H(Z)$ with N poles and M zeros.

$$H(Z) = \frac{A(Z)}{1 - B(Z)} = \frac{\sum_{i=0}^{M} a_i z^{-i}}{\sum_{i=1}^{N} b_i z^{-i}} \qquad (1)$$

Its equivalent difference equation is expressed as

$$y(n)=\sum_{i=0}^{M} a_i(n)u(n-i)+\sum_{i=1}^{N} b_i(n)y(n-i) \quad (2)$$

Eqs.(1) and (2) represent the realization of the unstrained direct-form IIR filter. This realization has the advantage of simple implementation and analysis. However it suffers potential shortcomings like instability, local optimum and slow convergence. One way to reduce such undesirable behavior is to use the Laguerre model. The transform function of a Laguerre model is

$$H(Z)=\sum_{i=0}^{M} w_i L_k(z) \quad (3)$$

where,

$$L_k(z)=L_0(z)(L(z))^k, L_0(z)=\frac{\sqrt{1-\alpha^2}}{1-\alpha z^{-1}} \text{ and } L(z)=\frac{z^{-1}-\alpha}{1-\alpha z^{-1}} \quad (4)$$

The parameter α denotes pole location which can be optimized according to some criteria. The most important characteristics of Laguerre filters are guaranteed stability and unique global minimum.

3. ADAPTIVE IIR ALGORITHM

This section presents an analysis of the output error method (OE), the equation error method (EE) and gradient adaptive Laguerre lattice (GALL) algorithm, which are the best known adaptive algorithms applicable to adaptive IIR filtering. In this paper, the OE and EE are implemented using unstrained direct form IIR filters and the GALL is performed for the Laguerre lattice filters to explore the individual properties of different filtering methods to be discussed in this section. The simulation results will be depicted in section 4.

3.1 The Output Error Method

The basic idea behind the output error method is to minimize the mean square value of the output error signal, which is defined by the difference between the actual system output and the adaptive filter output, to update the weight vector of the filter. Fig.3 shows the block algorithm of the OE method for system identification.

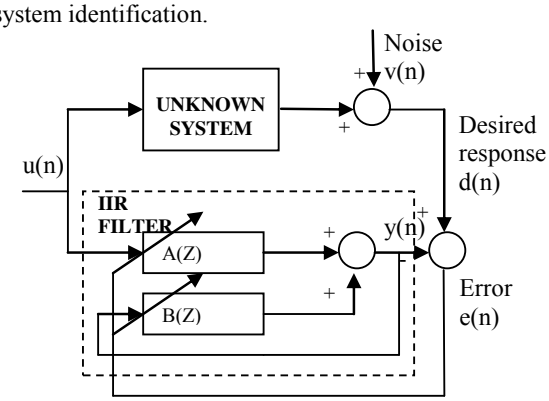


Fig.3 System identification illustration: output error method

Define the weight vector $\hat{\mathbf{w}}(n)$ as

$$\hat{\mathbf{w}}(n)=[a_0(n)\, a_1(n)\ldots a_M(n)\, b_1(n)\, b_2(n)\cdots b_N(n)]^T \quad (5)$$

The output error is

$$e_o(n)=d(n)-y(n) \quad (6)$$

Then the $e_o(n)$ is minimized based on the cost function

$$\mathrm{J}(n)=1/2E[|e_o(n)|^2] \quad (7)$$

An approach to solve this minimization problem is the stochastic gradient algorithm

$$\hat{\mathbf{w}}(n+1)=\hat{\mathbf{w}}(n)-\mu\nabla\mathrm{J(n)} \quad (8)$$

Consider a real-valued system identification problem, for a direct form IIR filter, the formulation of the LMS algorithm can be derived as follows:

- Take the cost function as

$$\mathrm{J}(n)=e_o(n)^2 \quad (9)$$

- Define the gradient vector as

$$g(n)=\nabla J(n)=\partial J(n)/\partial\hat{w}(n) \quad (10)$$

then the jth element is

$$g_j(n)=e_o(n)\partial e_o(n)/\partial\hat{w}_j(n)=-e_o(n)\partial y(n)/\partial\hat{w}_j(n) \quad (11)$$

with

$$\frac{\partial y(n)}{\partial\hat{w}_j(n)}=\begin{cases} u(n-j)+\sum_{i=1}^{N} b_i(n)\dfrac{\partial y(n-i)}{\partial a_j(n)} & \forall\hat{w}_j(n)=a_j(n) \\ y(n-j)+\sum_{i=1}^{N} b_i(n)\dfrac{\partial y(n-i)}{\partial b_j(n)} & \forall\hat{w}_j(n)=b_j(n) \end{cases}$$

- By assuming the filter adapts at a slow enough rate for small values of N and M, that is so called small step approximation, we have

$$\frac{\partial y(n-i)}{\partial a_j(n)}=\frac{\partial y(n-i)}{\partial a_j(n-i)} \quad \text{and} \quad \frac{\partial y(n-i)}{\partial b_j(n)}=\frac{\partial y(n-i)}{\partial b_j(n-i)}$$

Let $\alpha_i(n)=\dfrac{\partial y(n)}{\partial a_j(n)}$ and $\beta_i(n)=\dfrac{\partial y(n)}{\partial b_j(n)}$

- The weight updated equation is given by

$$\begin{aligned}\hat{\mathbf{w}}(n+1)&=\hat{\mathbf{w}}(n)-\mu\nabla\mathrm{J(n)}\\ &\approx\hat{\mathbf{w}}(n)+\mu e_0(n)\gamma(n)\end{aligned} \quad (12)$$

where $\gamma(n)$ is the regressor vector expressed by

$$\gamma(n)=[\alpha_0(n)\ \alpha_1(n)\ldots\alpha_M(n)\ \beta_1(n)\ \beta_2(n)\cdots\beta_N(n)]^T$$

Follow the same theme, the weight updated equation using NLMS can be formulated accordingly

$$\hat{\mathbf{w}}(n+1)\approx\hat{\mathbf{w}}(n)+\frac{\mu}{\delta+\|\gamma(n)\|^2}\gamma(n)e_0(n) \quad (13)$$

The NLMS algorithm have the advantages over the LMS method in that it provides more flexibility for filtering implementation, in addition it can converge faster and reduce the instable problem by choosing proper learning parameters μ and δ.

There are two important properties of the output error method need to be emphasized: the likehood of instability and the possibility of local minima. The first property implies that the stability checking is required during the adaptive filtering process. Some research effort was spent in order to avoid this requirement, for example the equation error method, which will be discussed in the next subsection. The second can affect the overall convergence of the adaptive algorithm, and the existence of an unbiased global minimum solution even in presence of perturbation noise in the unknown system output signal. Extensive studied have conducted [1][2][3]. In [4] two importance consequences regarding the most general result about the unimodality of the

MSOE performance surface in cases of sufficient order identification are given:

Consequence 1 If $\hat{n}_d = n_d = 1$ and $\hat{n}_n \geq n_n \geq 1$, then there is only one equilibrium point, which is the global minimum.

Consequence 2 If the input u(n) is a white noise, the orders of the adaptive filter are strictly sufficient ($\hat{n}_d = n_d$ and $\hat{n}_n = n_n$, $\hat{n}_n - n_d + 1 \geq 0$), then there is only one equilibrium point, which is the global minimum.

where,

$\hat{n}_d / n_d =$ the order of the denominator of filter/plant

$\hat{n}_n / n_n =$ the order of the numerator of filter/plant

3.2 The Equation Error Method

Fig.4 is the block algorithm of the EE method for system identification. Compare Fig.4 and Fig.3, it can be seen that the only difference between the OE and EE is that during the training of the EE the desired response d(n) is substituted for the actual filter output y(n) in the second term on the right hand side of Eq.(2).

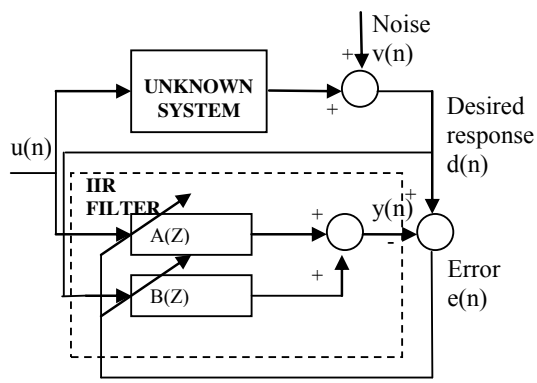


Fig.4 System identification illustration: equation error method

To introduce the EE method, consider the weight vector

$$\hat{\mathbf{w}}(n) = [a_0(n)\, a_1(n) \ldots a_M(n)\, b_1(n)\, b_2(n) \cdots b_N(n)]^T$$

and the regressor

$$X(n) = [u(n) \ldots u(n-M)\, d(n) \cdots d(n-N)]^T \quad (14)$$

The equation error is

$$e_e(n) = d(n) - X^T(n)\hat{w}(n) \quad (15)$$

From the equations above, it is readily to obtain the gradient version update equation following the same way in OE method.

The main characteristic of the equation error algorithm is the unimodality of the performance surface due to the linear relationship existent between the equation error signal and the adaptive filter coefficients. This property, however, comes along with the drawback of biased solution in the presence of a disturbance signal.

3.3 Gradient Adaptive Laguerre Lattice Algorithm

The reasons for employing the Laguerre Lattice realization in the real application are

- Guaranteed stability
- Unique global minimum
- Small number of parameters
- Good numerical behavior and low computer complexity similar to adaptive FIR lattice structure
- Relative insensitivity to eigenvalue spread of the input data autocorrelation matrix

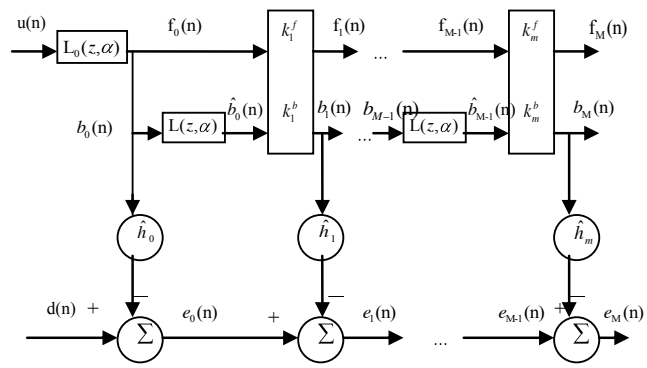


Fig.5 Laguerre lattice filter for joint-process estimation

Reffering to the Laguerre lattice filter shown in Fig.5, by adding the laguerre lattice stage, the forward $f_m(n)$ and backward $b_m(n)$ linear prediction errors are introduced. Using the definition of $L_0(Z)$ in Eq.(4) yields

$$f_0(n) = b_0(n) = L_0(Z)u(n) = \alpha b_0(n-1) + \sqrt{1-\alpha^2}\,u(n) \quad (17)$$

$f_m(n)$ and $b_m(n)$ can be calculated from $f_{m-1}(n)$ and $\hat{b}_{m-1}(n)$:

$$\begin{bmatrix} f_m(n) \\ b_m(n) \end{bmatrix} = \begin{bmatrix} 1 & -k_m^f(n) \\ -k_m^b(n) & 1 \end{bmatrix} \begin{bmatrix} f_{m-1}(n) \\ \hat{b}_{m-1}(n) \end{bmatrix} \quad (18)$$

where,

$$k_m^f(n) = \frac{E[f_{m-1}(n)\hat{b}^*_{m-1}(n)]}{E\left|\hat{b}_{m-1}(n)\right|^2}, \quad k_m^b(n) = \frac{E[\hat{b}_{m-1}(n) f^*_{m-1}(n)]}{E\left|f_{m-1}(n)\right|^2} \quad (19)$$

If the input signal u(n) is stationary, then

$$\hat{b}_{m-1}(n) = L(Z)b_{m-1}(n) \quad \Rightarrow$$

$$\hat{b}_{m-1}(n) = b_{m-1}(n-1) + \alpha(\hat{b}_{m-1}(n-1) - b_{m-1}(n)) \quad (20)$$

Based on the discussed concepts, a stochastic gradient (LMS-like) algorithm is given in [5] by minimizing the cost function

$$J_m = E[f_m^2(n) + b_m^2(n)] \quad (21)$$

The update equation for $\hat{h}_m(n)$ is

$$\hat{h}_m(n+1) = \hat{h}_m(n) + \frac{\hat{\mu}}{\|b_m(n)\|^2} b_m(n) e_m(n) \quad (22)$$

with

$$\|b_m(n)\|^2 = \|b_{m-1}(n)\|^2 + b_m^2(n) \quad (23)$$

4. NUMERICAL EXPERIMENTS

In this section, some numerical simulations are presented to illustrate some of the properties associated with the algorithms discussed previously. The system identification configurations have been selected for experiments, three examples are carried out. The first one is the basic one for exhibition of

the convergence behaviors of different algorithms including OE, EE and GALL; The second one is used to illustrate the coefficient sensitivity problem of IIR filters; The third is for evaluating the capability of IIR filters to track a time vary ARMA model of which the coefficients are subject to random perturbations.

4.1 Example 1

The unknown system is a second order ARMA process presented by

$$y(n) = 2u(n) + 1.5u(n-1) - 0.5y(n-1) - 0.25y(n-2) + v(n) \quad (24)$$

where, $v(n)$ is an additive noise at its output.

The filter has the following transfer function:

$$H(Z) = \frac{a_0 + a_1 z^{-1}}{1 - b_1 z^{-1} - b_2 z^{-2}} \quad (25)$$

Fig.6 through Fig.15 in the appendix show the simulation results.

The performance of OE and EE method: The weight tracing curves using OE method are in Figs.6 and 7, the corresponding EE method results are in Figs.8 and 9. It can be seen that the estimation values of the coefficients can nicely approach the actual value, with the suitable selections of the learning parameters μ and σ. Meanwhile, the results indicate the convergence using LMS has the problems of long convergence time, and noise amplification which may result in unstable behavior. To improve the performance with respect to these two issues the NLMS algorithm can be applied because NLMS can produce a faster and smoother convergence.

Since the order of the filter and the system are the same, which satisfies the condition of the **consequence 2** in section 3.1. No local minimum error surface is observed for this example when the OE method is implemented.

The performance of Gradient adaptive Laguerre lattice algorithm: Refer to the ensemble average learning curves and the mean values of the errors in Fig.12 and recall those of FIR in Fig.1. The GALL method can achieve the prescribed modeling error with significant less taps compare to FIR counterparts, and its behavior and computational cost are somehow similar to those for FIR. For example, from the Fig.12 it can be noted the estimation errors are significantly decreased when the order of the filter is increased form 2 to 3. but after the order reaches 4, the performance of the GALL can not be improved effectively by means of adding its order. On the other hand, compare the learning curves of GALL to those of OE and EE displayed in Figs.10 and 11. It reveals that the GALL has faster convergent speed but larger estimation error.

The performance of OE and EE under noise: The Fig.13 and Fig.14 present the weight tracing curves under different variances (0.01 and 0.25) of noises with the OE and EE method, respectively. Fig.13 tells us the final solutions using OE method under different sizes of white noises are the same. So the stable error of OE will not be affected by the additive white noise. However, the curves in Fig.14 show that in the EE case, the final solutions deviate to each other with a bias. The degree of the bias depends on the size of the variance of the disturbance.

The performance of OE and EE under white and colored noise: Fig.15 shows that the weight tracing curves of IIR under the white and colored noise disturbances. It illustrates that the performance are similar under these two different types of noises.

4.2 Example 2

To further inspect the IIR filter behavior, we now repeat the experiments for another stationary ARMA process expressed by

$$y(n) = 2u(n) + 1.5u(n-1) - 0.8y(n-1) - 0.7y(n-2) + v(n) \quad (26)$$

The filter has the same structure as Eq.(25). The objective is to see if the filter performance will change with the changes in the coefficients of the unknown system. The simulation results are exhibited in Fig.16 to Fig.18.

Fig.16 and 17 shows the OE method result, the corresponding simulation results of EE are omitted because they give similar information to OE. It can be seen for this particular example, the LMS based OE and EE method do not work anymore. The tracing curves can not converge to the actual value even with a very small step size, which means the filter is unstable. With the NLMS algorithm, it may still work if smaller learning parameters are chosen, however, compare it to example1, the performance is greatly degraded. As to the GALL algorithm, the learning curves in Fig.18 demonstrate that it ensures the stability. Compare to the results of example 1 in Fig. 12, it illustrates that with the same filter order, the estimation error of example 2 is much bigger than that of example 1, which implies if we want the same estimation accuracy, this filter in example 2 must be higher order, thus more complex structure. All this results show the dependence of the IIR filter performance on the system coefficients.

4.3 Example 3

The intention of this experiment is to examine the capability of the IIR filter to track a time varying ARMA process. The unknown system is modeled by

$$y(n) = a_1 y(n-1) + a_2 y(n-2) + a_3 u(n) + a_4 u(n-1) + v(n) \quad (27)$$

The coefficients are subject to random perturbations given by

$$a_k(n+1) = a_k(n) + w_k(n) \quad (28)$$

where, $w_k(n)$ are white noise and independent of $v(n)$.

The tracking curves are shown in Fig.19 to 22. Obvious bias and time delay are observed which means the IIR filter can not genuinely estimate the time vary system which are perturbed by the noise.

5. CONCLUSIONS

The concepts of IIR filter are reviewed and different methods are discussed. Through the extensive computer simulations, various IIR filters are studied.

Based on the natures of convergences shown in simulations, the properties of the studied filtering methods can be seen clearly. Each method has its own advantages/disadvantages. The potential shortcomings of IIR filters are instability, local optimum, low convergence rate, sensitive to initial condition and coefficient, bias under noise etc. To apply the IIR filters in the specific applications, one has to consider all available information to determine the most appropriate algorithm.

6. REFERENCES

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APPENDIX

FIGURES FOR SIMULATION RESULTS

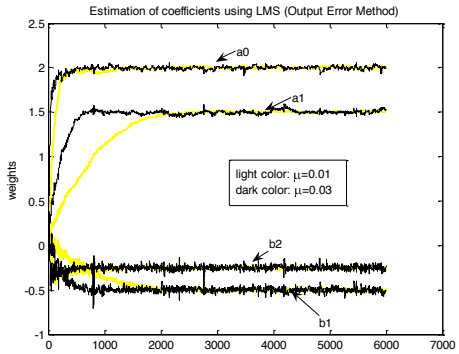


Fig.6 Output error method: LMS algorithm

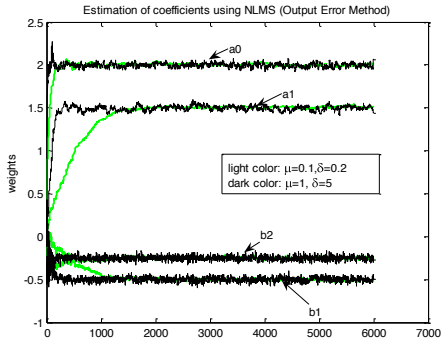


Fig.7 Output error method: NLMS algorithm

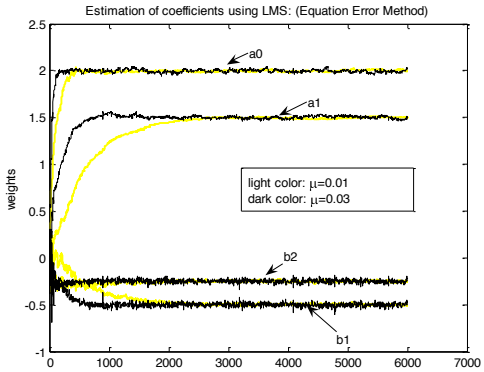


Fig.8 Equation error method: LMS algorithm

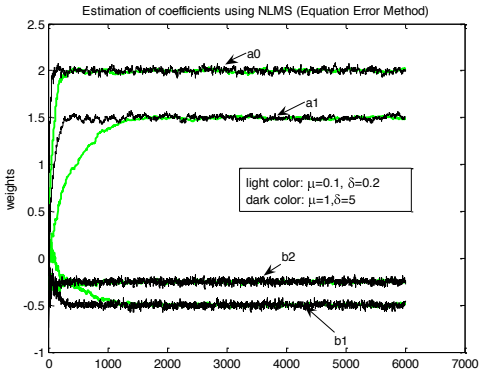
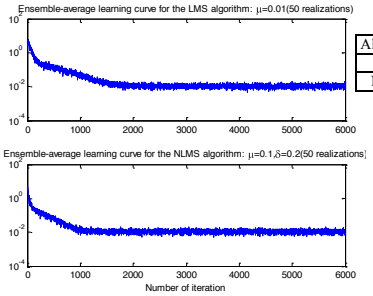
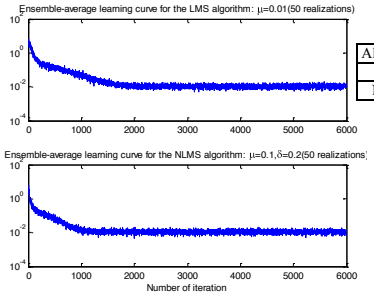


Fig.9 Equation error method: NLMS algorithm



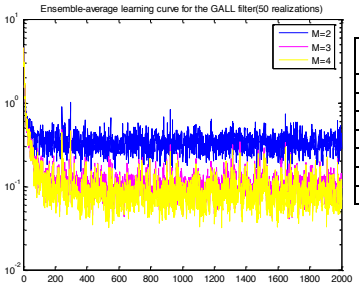
Algorithm	Mean	Std
LMS	0.075	0.358
NLMS	0.039	0.199

Fig.10 Learning curve: output error method



Algorithm	Mean	Std
LMS	0.074	0.342
NLMS	0.057	0.284

Fig.11 Learning curve: equation error method



Filter order	Mean	Std
1	0.8589	0.2286
2	0.8852	0.2291
3	0.3504	0.1935
4	0.1257	0.1970
5	0.1093	0.1973
6	0.1077	0.1977
10	0.1031	0.1980

Fig.12 Learning curve: Gradient adaptive Laguerre lattice algorithm

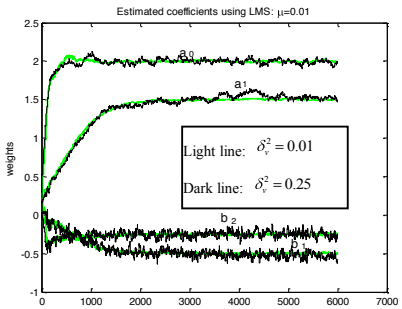


Fig.13 Output error method: Tracing performance under different size of noise

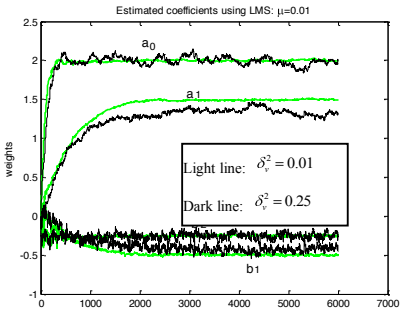


Fig.14 Equation error method: Tracing performance under different size of noise

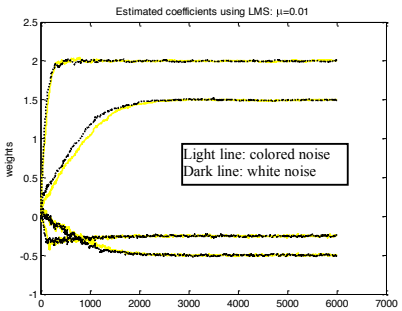


Fig.15 Comparison of performance under white and colored noise

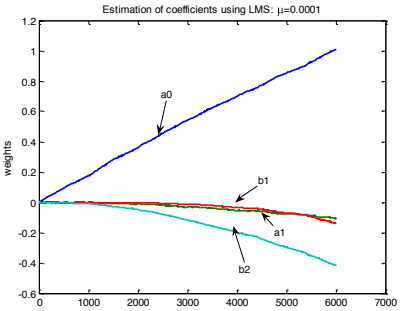


Fig.16 Output error method for example 2: LMS

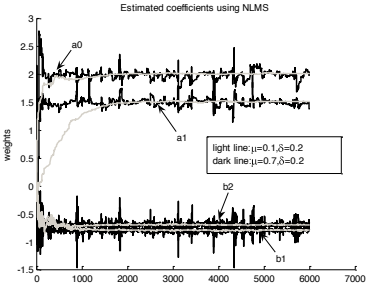
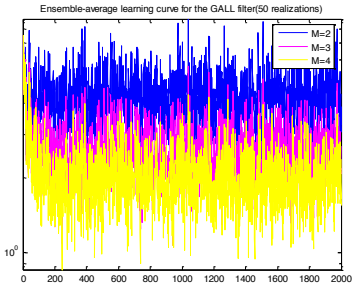


Fig.17 Output error method for example 2: NLMS



Filter order	Mean	Std
1	5.0228	1.0706
2	4.3182	0.9608
3	2.6072	0.6601
4	2.0343	0.5861
5	2.0209	0.5849
6	1.5177	0.5558
7	1.1081	0.5503
8	1.0861	0.5511
9	0.9994	0.5543
10	0.8186	0.5686

Fig.18 GALL algorithm for example 2

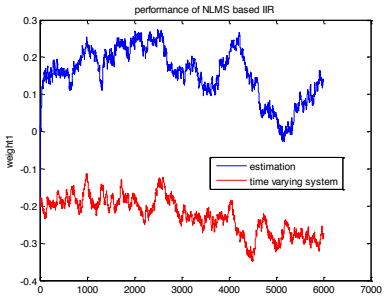


Fig.19 Tracing performance for example 3: weight 1

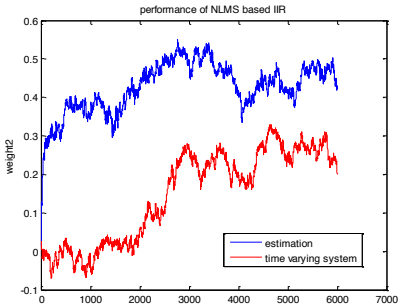


Fig.20 Tracing performance for example 3: weight 2

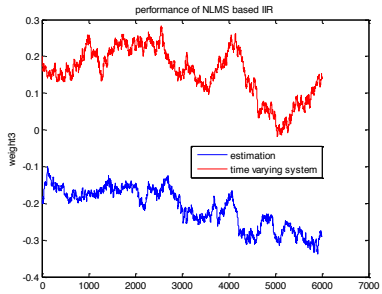


Fig.21 Tracing performance for example 3: weight 3

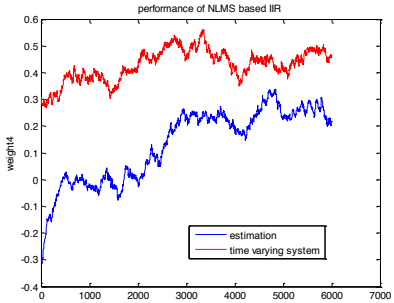


Fig.22 Tracing performance for example 3: weight 4